

House Price Prediction Project Report

1. Project Description

The purpose of this project is to predict house sale prices using the Kaggle House Prices dataset. The response variable is **SalePrice**. Linear regression is used to study which housing features are associated with price and to generate predictions for the test dataset.

2. Data Description

The training data contains **1460 rows and 81 columns**, while the test data contains **1459 rows and 80 columns**. Each row represents one home sale. The dataset includes both numerical and categorical housing characteristics such as lot size, neighborhood, quality ratings, basement area, and garage information. The variable definitions provided by the dataset describe variables such as 'MSSubClass', 'MSZoning', 'Neighborhood', 'OverallQual', and 'KitchenQual'.

3. Data Cleaning

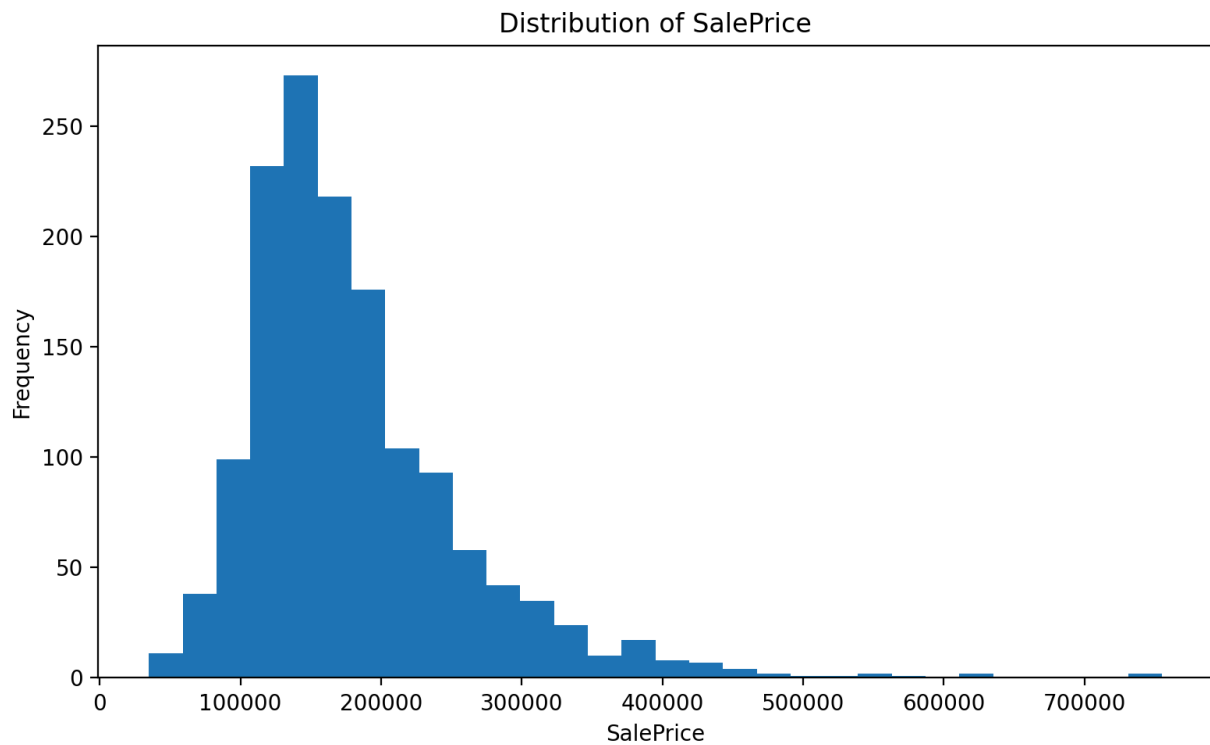
To match the professor's starter notebook, the following columns were dropped because they were identifiers or had substantial missing values:

'Id', 'LotFrontage', 'Alley', 'MasVnrType', 'MasVnrArea', 'BsmtQual', 'BsmtCond', 'BsmtExposure', 'BsmtFinType1', 'BsmtFinType2', 'FireplaceQu', 'GarageType', 'GarageYrBlt', 'GarageFinish', 'GarageQual', 'GarageCond', 'PoolQC', 'Fence', and 'MiscFeature'.

After dropping these columns:

- object variables were converted to categorical variables
- the remaining missing rows were removed from the training set
- one row was lost, leaving **1459 complete training observations**

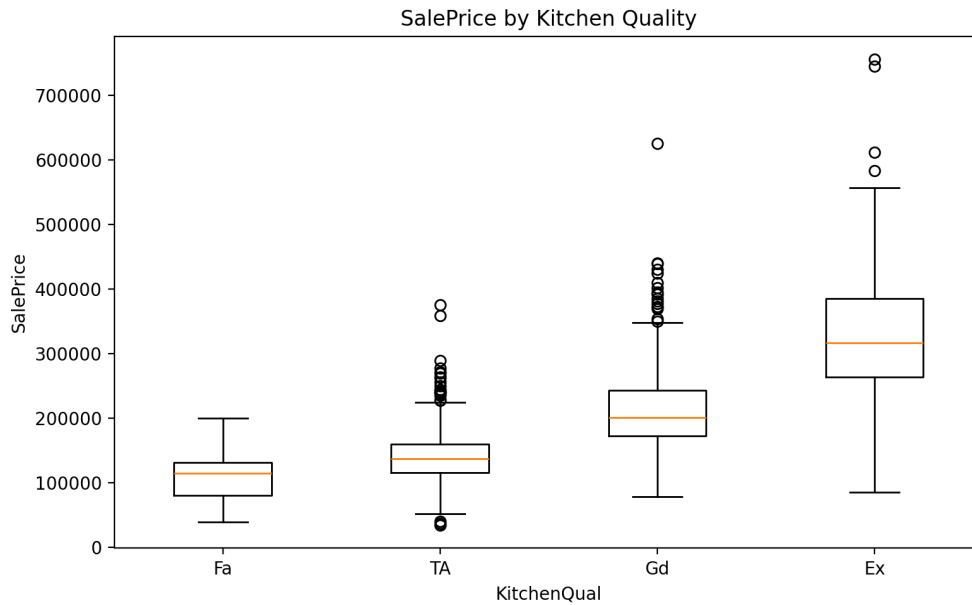
4. Data Visualization



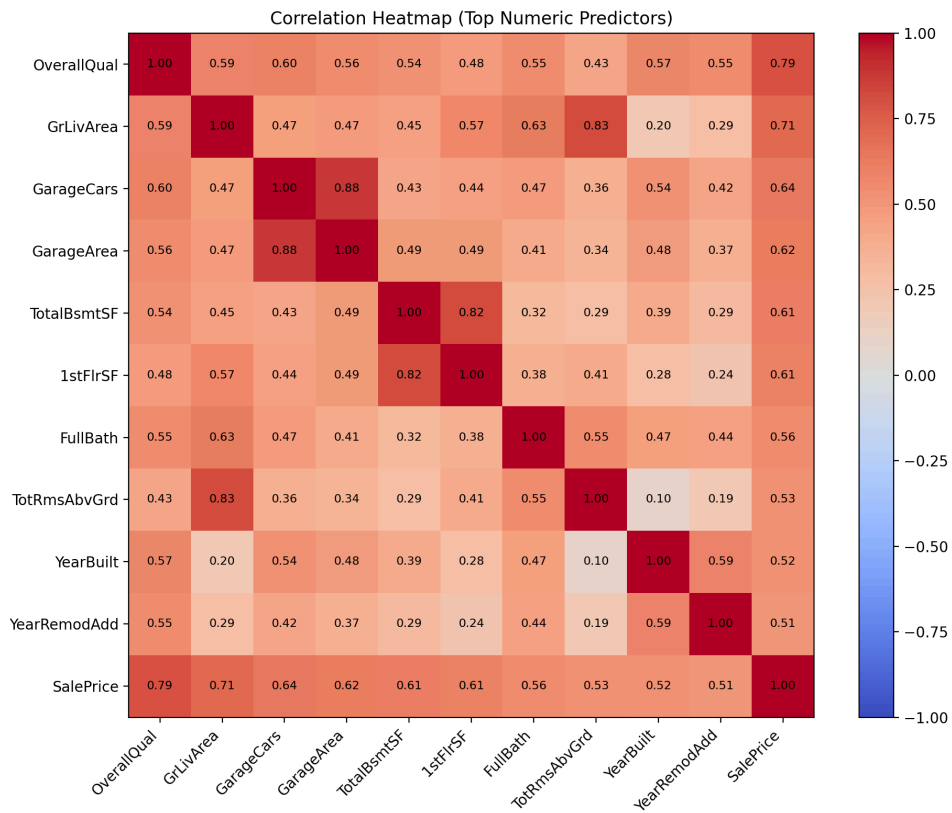
The histogram of `SalePrice` shows a right-skewed distribution, meaning most houses are in a moderate price range while a smaller number are very expensive.

The correlation analysis showed that the strongest numerical predictors of `SalePrice` were:

- `OverallQual`
- `GrLivArea`
- `GarageCars`
- `GarageArea`
- `TotalBsmtSF`



The boxplot for `KitchenQual` shows that better kitchen quality is generally associated with higher sale prices.



The correlation heatmap shows the relationships between key numerical variables and SalePrice. Variables such as OverallQual, GrLivArea, GarageCars, GarageArea, and

TotalBsmtSF show strong positive correlations with SalePrice, indicating they are important predictors. These variables were therefore included in the regression models.

5. Regression Models

Three models were built:

Full Model

`SalePrice ~ OverallQual + GrLivArea + GarageCars + GarageArea + TotalBsmtSF + FullBath + TotRmsAbvGrd + YearBuilt + YearRemodAdd + KitchenQual + Neighborhood + ExterQual + BsmtFinSF1 + LotArea + Fireplaces`

Reduced Model

`SalePrice ~ OverallQual + GrLivArea + GarageCars + TotalBsmtSF + YearBuilt + KitchenQual + Neighborhood`

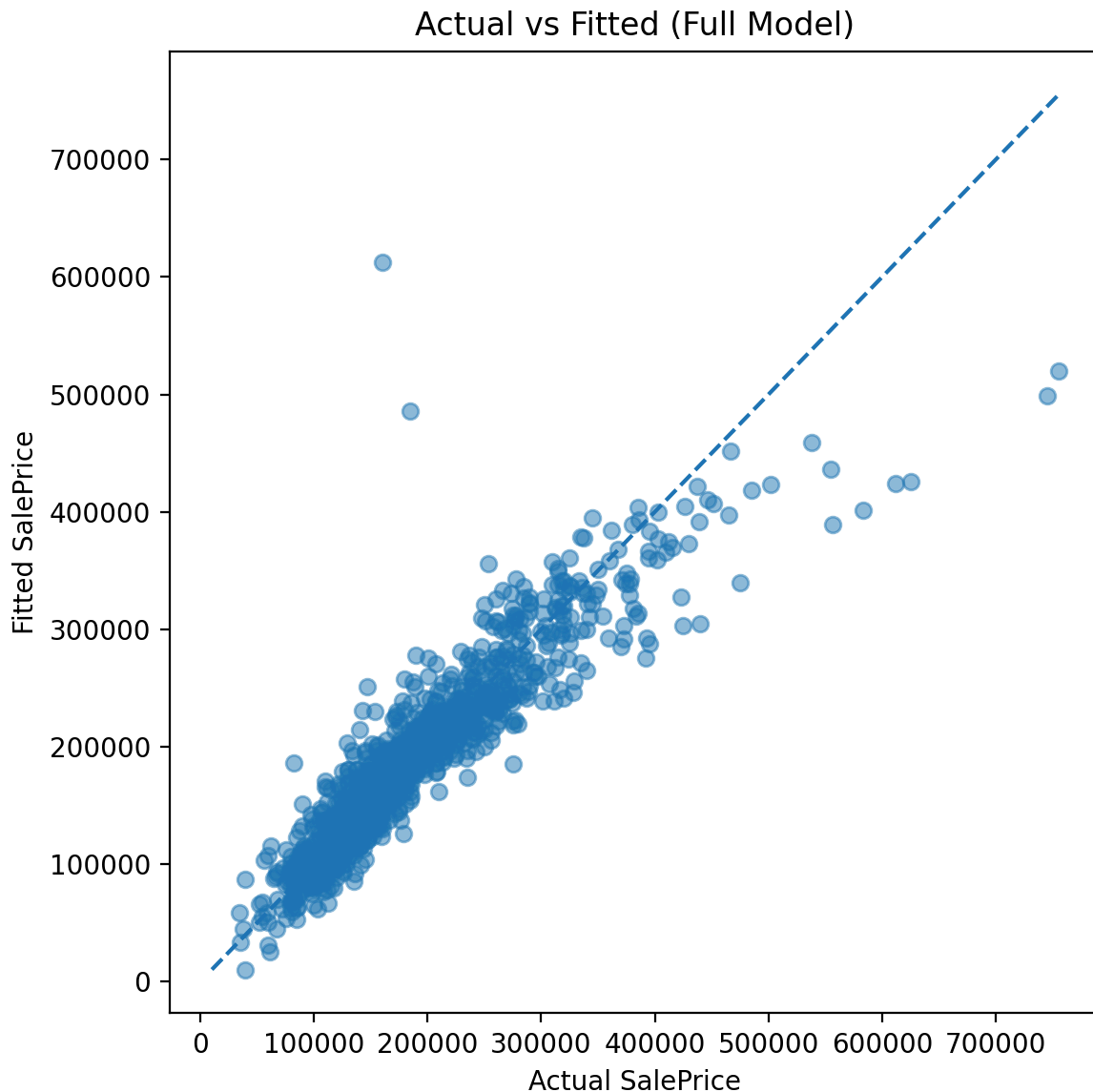
Interaction Model

`SalePrice ~ GarageCars + TotalBsmtSF + YearBuilt + KitchenQual + Neighborhood + GrLivArea \* OverallQual`

6. Model Comparison

Model	R ²	Adjusted R ²	AIC	BIC
Full	0.8423	0.8376	34455.19	34682.46
Reduced	0.8254	0.8215	34583.25	34757.67
Interaction	0.8318	0.8279	34530.92	34710.63

7. Final Model Choice



The actual vs fitted plot shows a strong positive linear relationship between observed and predicted values, indicating that the model provides a good fit to the data. While there is some spread, the model captures the overall trend in house prices effectively.

The **full model** was selected as the final model because:

- it had the **highest R^2**
- it also had the **lowest AIC and BIC**
- it explained the greatest proportion of variation in house prices among the three models

An R^2 of **0.8423** means that the model explains about **84.23%** of the variability in house sale prices.

8. Interpretation of Important Coefficients

From the final model, several variables were especially important:

- **OverallQual** had a strong positive coefficient, meaning higher overall material/finish quality is associated with higher sale price.
- **GrLivArea** had a positive coefficient, meaning larger above-ground living area is associated with higher sale price.
- **YearRemodAdd** had a positive coefficient, suggesting more recently remodeled homes tend to sell for more.
- **Fireplaces** had a positive coefficient, indicating that homes with more fireplaces tend to have higher sale prices.
- Some categories of **KitchenQual**, **ExterQual**, and **Neighborhood** were also significant, showing that quality and location matter.

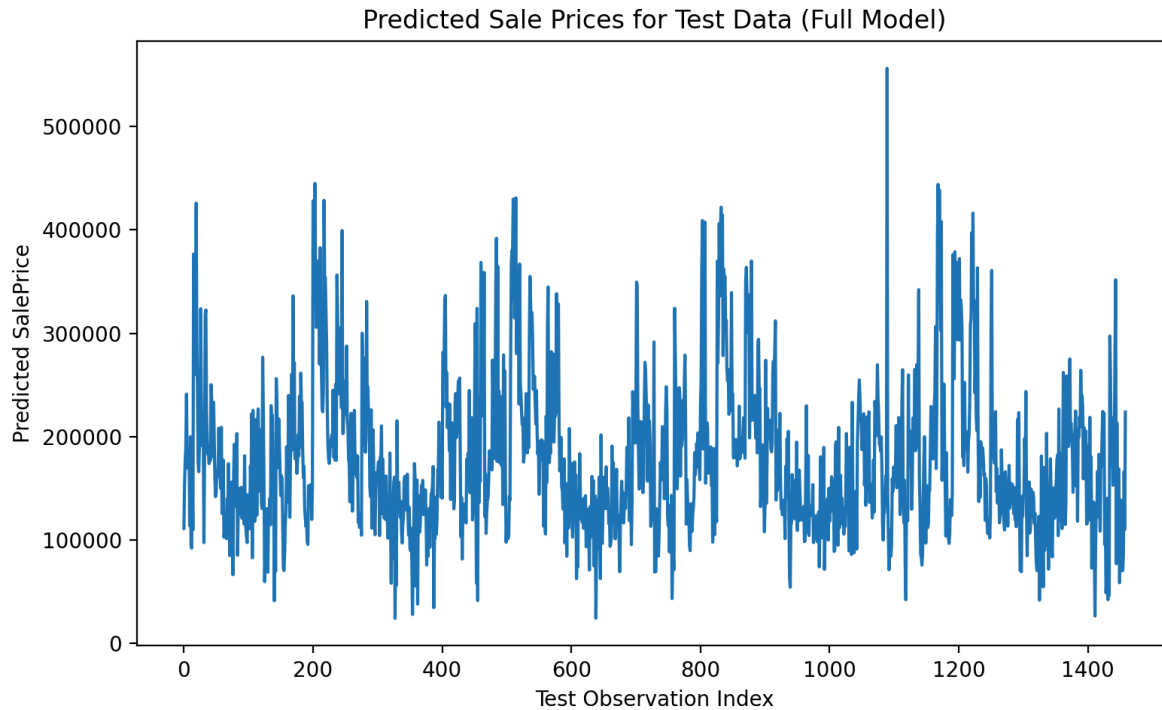
Example interpretation:

Holding other variables constant, a one-unit increase in OverallQual is associated with an increase of approximately \$11,821 in the predicted sale price.

Holding other variables constant, a one-square-foot increase in GrLivArea is associated with an increase of approximately \$35 in the predicted sale price.

Holding other variables constant, each additional garage space (GarageCars) is associated with an increase of approximately \$10,353 in the predicted sale price.

9. Prediction on Test Data



The final full model was used to predict house prices for the 1459 observations in the test dataset. Before prediction, missing values in the test set were filled using:

- the **median** for numerical variables
- the **mode** for categorical variables

The prediction file was saved in Kaggle-submission format with columns:

- `Id`
- `SalePrice`

10. Conclusion

This project used linear regression to model house prices based on structural, quality, and location-related variables. Exploratory analysis identified key predictors such as OverallQual, GrLivArea, and GarageCars. After comparing three models, the full model was selected because it provided the best overall fit, with the highest R^2 and lowest AIC/BIC. The final model explained approximately 84% of the variability in house prices and was used to generate predictions for the test dataset.